

**Problem 1 (20 Points):** Let  $\theta > 0$  be an unknown parameter, and let  $X = (X_0, X_1, \dots, X_n)$ , where  $X_0 \sim \text{Poisson}(\theta)$  and for any  $i \in [n]$ ,

$$X_i | (X_0, X_1, \dots, X_{i-1}) \sim \text{Poisson}(\theta X_{i-1}).$$

- i) (10 Points) Find the MLE of  $\theta$  based on the observation  $X$ .
- ii) (10 Points) Compute  $I(\theta)$  with respect to the observation  $X$ .

**Problem 2 (20 Points):** Let  $X = (X_1, \dots, X_n)$ , where

$$X_i \stackrel{iid}{\sim} f_\theta(x_i) = \frac{1}{2} \exp(-|x_i - \theta|),$$

and  $\theta > 0$  is the unknown parameter. Find an MLE of  $\theta$ . (Hint: Consider the cases where  $n$  is even and odd separately and find the MLE in terms of the order statistics.)

**Problem 3 (20 Points):** Given a fixed known integer  $r > 1$ , let  $X = X_{i,j} : i \in [n], j \in [r]$  such that

$$X_{i,j} \sim \mathcal{N}(\mu_i, \sigma^2)$$

and  $X_{i,j}$ 's are mutually independent. The unknown parameters are  $\theta = (\mu_1, \dots, \mu_n, \sigma^2)$ , where  $\mu_i$  can be any real number for all  $i \in [n]$  and  $\sigma^2 > 0$ .

- i) (10 Points) Find the MLE of  $\theta$ .
- ii) (10 Points) Is the MLE asymptotically unbiased for  $\sigma^2$  as  $n \rightarrow \infty$ ? Can you explain, intuitively, why or why not?

**Problem 4 (20 Points):** Let  $X = (X_1, \dots, X_n)$ , where

$$X_i \stackrel{iid}{\sim} \theta f_1(x_i) + (1 - \theta) f_2(x_i).$$

Here,  $f_1$  and  $f_2$  are two different known density functions, and  $\theta \in [0, 1]$  is the unknown parameter.

- i) (10 Points) Find a sufficient and necessary condition for which the score equation

$$\frac{d}{d\theta} \log f_\theta(x) = 0$$

has a unique solution. Show that if the score equation has a unique solution, it is the MLE.

ii) (10 Points) Find the MLE when the score equation has no solution.

**Problem 5 (40 Points):** You have two biased coins (Coin A and Coin B) with unknown probabilities of landing on heads ( $\theta_A$  and  $\theta_B$ ). You pick one coin at random, flip it 10 times, and record the results (e.g., (5H, 5T)). You repeat this five times for a total of 50 flips. You do not know which coin was used for each set of 10 flips. This “which coin” information is the latent variable ( $U_i \in \{A, B\}, i = 1, 2, 3, 4, 5$ ). The goal here is to use the EM algorithm to estimate the true bias ( $\theta_A$  and  $\theta_B$ ) for each coin.

i) (20 Points) Find an explicit expression for the posterior distribution and the expectation

$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x) \quad \text{and} \quad \mathbb{E}_{q^{(t)}} [\log f_{\theta}(x, U)]$$

in the E-step and the update rule

$$\theta^{(t)} = \arg \max_{\theta} \mathbb{E}_{q^{(t)}} [\log f_{\theta}(x, U)]$$

in the M-step.

ii) (20 Points) Given that the observations for the five trials are:

Trial 1 (5H, 5T), Trial 2 (9H, 1T), Trial 3 (8H, 2T), Trial 4 (4H, 6T), Trial 5 (7H, 3T),

write a compute program to find the MLE of  $\theta = (\theta_A, \theta_B)$ . Plot the posterior probabilities  $p_{\theta^{(t-1)}}(u_i = A|x), i = 1, 2, 3, 4, 5$  and the estimates  $(\theta_A^{(t)}, \theta_B^{(t)})$  as a function of the number of iterations  $t$ .